FLATNESS OF A MORPHISM OF ALGEBRAIC SPACES EQUALS FLATNESS OF THE ASSOCIATED MORPHISM OF RINGED SITES

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Editorial note. All delivered mathematical text is the author's own, copied from the pinned Stacks Project chapter file spaces-morphisms.tex; no statement or proof step was rewritten, reordered or removed.

Editorial note. The statement displays the morphism of ringed sites in multi-line notation; the proof unwinds the definition and reduces both sides to the same stalk criterion.

Editorial note. The three references of the counted spans are recorded as non-load-bearing navigation pointers; the argument that consumes them stays inside the counted proof span.

Editorial note. Preamble: the three native excerpt spans of sources/stacks-spaces-morphisms/preamble.tex (lines 8-13, 14-18 and 154-217). The chapter file itself carries no preamble text.

Allowed standard prerequisites (editorial).

- Flatness of a morphism of ringed sites as flatness of the structure sheaf module. used at proof lines 6234-6235. Fixes the meaning of the left hand side.
- Flatness of a module can be checked at stalks. Modules on Sites, Lemma sites-modules-lemma-check-flat-stalks, used at proof lines 6236-6237. Reduces the site-theoretic statement to stalks.
- Exactness of the stalk functor on the etale site. Properties of Spaces, Theorem spaces-properties-theorem-exactness-stalks, used at proof lines 6238-6239. Makes the stalk criterion available.
- Flatness of a morphism of algebraic spaces checked at points. Lemma lemma-flat-at-point-etale-local-rings (6162-6175), used at proof lines 6240-6241. Provides the comparison with the space-theoretic notion.

Source reference. Original lemma 04NG: focused-seeds/AGP/sources/stacks-snapshot-20261011/tags.txt, source lines 6079–6079. Navigation pointer into the Stacks Project; the standard fact it supplies is applied inside the counted proof span and the argument resting on it stays there.

Source reference. Original lemma 05VC: focused-seeds/AGP/sources/stacks-snapshot-20261011/tags.txt, source lines 7546–7546. Navigation pointer into the Stacks Project; the standard fact it supplies is applied inside the counted proof span and the argument resting on it stays there.

Source reference. Original theorem 04K5: focused-seeds/AGP/sources/stacks-snapshot-20261011/tags.txt, source lines 5961–5961. Navigation pointer into the Stacks Project; the standard fact it supplies is applied inside the counted proof span and the argument resting on it stays there.

Date: stacks-project:spaces-morphisms.tex:local-snapshot-20261012; GNU FDL 1.2 or later.

Lemma 0.2. *Let S be a scheme. Let $f : X \to Y$ be a morphism of algebraic spaces over S. Then f is flat if and only if the morphism of sites $(f_{small}, f^\sharp) : (X_{étale}, \mathcal{O}_X) \to (Y_{étale}, \mathcal{O}_Y)$ associated to f is flat.*

Proof. Flatness of $(f_{small}, f^\sharp)$ is defined in terms of flatness of $\mathcal{O}_X$ as a $f_{small}^{-1}\mathcal{O}_Y$-module. This can be checked at stalks, see Modules on Sites, Lemma 05VC and Properties of Spaces, Theorem 04K5. But we've already seen that flatness of f can be checked on stalks, see Lemma 04NG. □